

Martin Tembo

Lead Software Engineer | Full Stack & Systems Engineer

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Professional Summary

Full stack and systems engineer with 4+ years of professional experience (coding since 2017), currently Lead Software Engineer for a U.S. company. I ship complete products end to end — iOS and Android apps, desktop apps, web platforms, and the backends behind them — in TypeScript, Go, Python, and Rust. Recent work spans AI tutoring with RAG, tool calling, and real-time voice; payment and subscription systems designed for correctness under failure; and media pipelines for adaptive video streaming. Experienced working remotely with teams in the USA, Australia, and Bangladesh.

Technical Skills

Languages: TypeScript, JavaScript, Python, Go, Rust, Swift, SQL, GraphQL

Frontend & Mobile: React, Next.js, React Native, Expo, Tauri, SwiftUI, TailwindCSS, Apollo Client

Backend: Node.js, Bun, Hono, NestJS, Express.js, FastAPI, Gin, GraphQL, WebSockets

AI Engineering: LLM integration (Gemini, OpenAI), RAG with pgvector, tool calling, real-time voice (Gemini Live), AI agents, MCP, guardrails and cost controls

Data & Messaging: PostgreSQL, MongoDB, Redis, MySQL, Prisma, TypeORM, BullMQ, RabbitMQ, pg-boss

Cloud & DevOps: AWS (Lambda, S3), Fly.io, Cloudflare R2, Docker, Bazel, GitHub Actions, CI/CD, OpenTelemetry, Sentry, PostHog, Linux

Media & Real-Time: LiveKit, FFmpeg, HLS streaming, whisper.cpp, VoIP push / CallKit

System Design: Domain-Driven Design, modular monoliths, microservices, event-driven architecture, transactional outbox, idempotency, reconciliation

Work Experience

ATB Applications LLC (formerly ShaftFitters LLC) | USA (Remote)

Apr 2026 – Present

Lead Software Engineer

- Lead engineer on ShaftFitters, a golf shaft recommendation platform built from golfers' own swing data — owning backend, mobile, and web end to end. Live on the U.S. App Store.
- Built the iOS and Android apps in Expo and React Native, with swing-data visualisations in Skia and Reanimated.
- Architected the Hono REST API on AWS Lambda and Fly.io with PostgreSQL, Prisma, and Redis, plus BullMQ queues powering the recommendation engine, maintenance, and catalog sync.
- Implemented App Store and Google Play subscriptions via RevenueCat with webhook-driven entitlements, plus Stripe payments and pay-as-you-go plans.
- Built the Next.js web app and admin console, an affiliate product catalog with supplier purchase orders, and JWT/OTP auth with role-based access; ran production with Sentry, PostHog, and go-live database audits.

Tech: React Native, Expo, Next.js, TypeScript, Hono, PostgreSQL, Prisma, Redis, BullMQ, AWS Lambda, Fly.io, RevenueCat, Stripe, Sentry, PostHog

Homiee | Australia (Remote)

Feb 2025 – 2026

Full Stack Developer

- Architected core systems for an Australian real estate platform, including property search with sub-100ms latency and agent-facing dashboards for property management and analytics.
- Engineered a real-time AI chat system enabling Q&A between users and OpenAI-based agents.
- Developed a Python computer-vision pipeline that detects buildings in 4K property videos, rendered as frame-synchronised, clickable HTML5 Canvas overlays.
- Cut detection payloads from 200MB+ to 2–4MB with whitespace stripping and Gzip, and built chunked 4K uploads with .ts streaming for unstable networks.

Tech: Next.js, React, Node.js, Express.js, Python, MongoDB, AWS, OpenAI, HTML5 Canvas

Backend & Systems Lead / Full Stack Developer

- Engineered NestJS backend systems supporting 100,000+ active job listings, with REST APIs for listings, contracts, and user workflows.
- Reduced database query times by 40% through advanced PostgreSQL indexing strategies.
- Designed secure authentication flows and role-based access control (RBAC).
- Built a Google Business Reviews aggregation system on Go (Gin) microservices, with real-time sync between external APIs and internal dashboards.
- Created responsive React interfaces for complex data visualisation.

Tech: NestJS, TypeScript, PostgreSQL, MongoDB, Golang, Gin, React, Docker, GCP

Shypass | Zambia**2022****Full Stack Developer**

- First professional role, building frontend and backend features across the product.

Tech: ReactJS, JavaScript, Express.js, MongoDB

Featured Projects

Tutorly-AF Academy (Live — tutorly-af.academy)

Learning platform for multi-country curricula, launching in Zambia: an Expo app for students, a Tauri desktop studio for tutors, a GraphQL backend, and a Rust media pipeline. Features live classes on self-hosted LiveKit, an AI tutor with RAG, tool calling and real-time voice, end-to-end encrypted chat, transcript-based video editing, and adaptive HLS streaming.

Tech: React Native, Expo, Tauri, Next.js, Bun, Hono, GraphQL, Rust, FFmpeg, LiveKit, Gemini, pgvector, PostgreSQL, Redis

Paycore (In development)

Payment and subscription platform for Zambia, built to stay correct under failure: a Go modular monolith with Lipila mobile money collections, idempotency keys, a transactional outbox, reconciliation, distributed locks, and auditable state machines, plus a native SwiftUI iOS app.

Tech: Go, PostgreSQL, Redis, RabbitMQ, OpenTelemetry, Bazel, Protobuf, OpenAPI, SwiftUI

PricePul (Live — pricepul.com)

B2B pricing intelligence SaaS that monitors competitor pricing pages 24/7 with AI-powered change detection and Slack/CRM alerts for sales teams.

Tech: Next.js, Hono, Prisma, PostgreSQL, Redis, TailwindCSS

AudienceAce (Live — audienceace.com)

AI-powered lead generation using NLP to scan social platforms (Reddit, LinkedIn) for high-intent buyer signals, with lead scoring and automated outreach.

Tech: AI Agents, NestJS, TypeORM, Python, RabbitMQ, Redis, PostgreSQL

Eurepraxis (In development)

Research operating system that connects evidence, contradictions, and knowledge gaps across scientific literature into testable research directions, with every claim traceable to a source passage.

Tech: Rust, Python, FastAPI, MCP, LLMs, MongoDB, Redis

Peptide Maxxing (Live — peptidemaxxing.com)

U.S. e-commerce store for research-grade peptides. Contributed dashboard analytics and other admin dashboard features in 2026.

Tech: Next.js, React